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323, Abid House, 1st Floor,
Opp. Lamington Road Police Station,
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Types of T&M devices

Broadly, the T&M products can be classified into products for general-purpose test and measurement; video test, measurement and monitoring; and communication diagnostics and network management.

The general-purpose test products include oscilloscopes, logic analysers, signal sources and spectrum analysers. Oscilloscopes are the primary general-purpose electronic debug tools. Logic analysers are debug tools used by design engineers to capture, display and analyse streams of digital data that occur simultaneously over many channels. Signal sources are general-purpose stimulus products primarily used in the design and manufacturing of electronic components, sub-assemblies and end products in a wide variety of industries. Spectrum analysers are primarily used in the design and manufacture of electronic components, sub-assemblies and end products that utilise digital RF technology.

Video measurement equipment are used to ensure delivery of the best possible video experience to the viewer, either through traditional analogue television, or through digital terrestrial, satellite, cable or broadband services. These include waveform monitors, video signal generators and compressed digital video test products. They are used in video equipment design and manufacturing, video content production, and video transmission and distribution.

Similarly, network management and diagnostic tools enable network equipment manufacturers and operators to develop, deploy, and manage mobile and fixed-line networks. Network management tools provide continuous performance management to optimise the service performance of an entire communications network. Diagnostic tools test and monitor the signaling protocols in next-generation communication networks and services.

to get to root cause as soon as possible. To do this, it is necessary to pinpoint the problem in the DUT. Being able to correlate data taken at the time of a failure at multiple points in the design, in multiple domains, is essential to providing rapid insight.

Finally, their budget hasn't changed. They can buy a spectrum analyser/VSA combination device, or an RTSA, but not both. All the functions of a high-performance spectrum analyser and vector signal analyser must be included in their troubleshooting and characterisation tool.

Embedded system design challenge. Today, we see a vast array of complex electronic circuits, or embedded system designs, in a wide range of applications—from low-cost to high-performance. The design engineers creating these embedded systems are faced with many of the same design challenges.

Today's designs often feature many different elements, including microcontrollers or field-programmable gate arrays (FPGAs), serial and parallel buses, analogue-to-digital converters (ADCs), digital-to-analogue converters (DACs) and power supplies, all in a single de-

sign. This creates debug challenges for the design engineers.

Designers must monitor a wide range of signals to troubleshoot system-level problems. They may need to trace data flow through their device, requiring them to look at several buses and time-correlate data packets. Or, they may need to troubleshoot a mixed-signal element like an ADC or DAC with both digital and analogue signals. Or, they may need to look at the modulation characteristics of their pulse width modulator. Looking at crosstalk between signals would be another example.

Troubleshooting complex embedded system designs often requires looking at several signals at one time. Tracing data flow through a design may require looking at two 3-wire buses. Troubleshooting an ADC may require looking at an 8-bit parallel bus and an analogue signal. These complex designs often require more than the four channels available on a traditional oscilloscope.

Designers also need more than just signal visualisation. Serial and parallel buses have become pervasive across applications. In a recent product re-